

SAHIL TOMAR

🌐 human-in-loop.dev

☎ (+91) 94578 18390

✉ dev.sahil.tomar@gmail.com

🌐 linkedin.com/in/dev-s-t

🐙 github.com/dev-S-t

Summary

AI Solutions Engineer building production AI systems, not prototypes. Leading TechieMaya's Voice AI vertical (VOAG): **1,000+ calls/day at sub-300ms p95 latency** on self-hosted LiveKit and SIP telephony, with multi-provider model routing and per-tenant cost tracking.

Comfortable shipping into constrained, real-world environments — client VMs, on-premise networks, strict NAT gateways — and staying accountable for the system long after go-live, from monitoring and debugging to cost tuning, without a dedicated ops team.

Equally hands-on with multi-agent systems (Google ADK, LangGraph), RAG over private data, and real-time speech pipelines in **Python and Go** on GCP.

Work Experience

TechieMaya

AI Solutions Engineer

Apr. 2025 – Present

Remote (Dubai, UAE)

- **VOAG — Enterprise Voice AI SaaS** Mr. LADs
 - **Lead the Voice AI vertical:** scaled to **1,000+ calls/day at sub-300ms p95 latency**; replaced persistent WebSockets with an *async polling layer* waking Cloud Run workers on demand, cutting infrastructure cost **30%**.
 - **Hybrid cloud SIP:** engineered a UAE telephony bridge (SIM → modem → SIP → LiveKit) deployed *inside a client's own VM behind a strict NAT gateway*; self-hosted LiveKit across India and UAE with tenant-aware routing and **GitHub Actions**-driven CI/CD.
 - **Agentic workflows & multi-provider routing:** built a tenant-aware tool registry over OAuth 2.0 for live calendar scheduling, omnichannel messaging, and mid-call human handoff; added *fire-and-forget RAG* for long-document reference with no conversational dead air, routed across Gemini Live, Sarvam, and Ultravox for cost/language optimisation, with **Cartesia** as the primary TTS layer (plus Fish Audio) for brand-matched *voice cloning*.
- **WhatsApp Dispatch Automation** — B2B, luxury ground transport (Dubai)
 - Built a Go gateway (*whatsmeow*) for group-chat messaging unsupported by Meta's official API, paired with a **Google ADK multi-agent system** (orchestrator, booking, support agents) resolving flight, maps, fleet-availability and fare data via tools — cutting booking turnaround **from ~30 minutes to under 1 minute**; extended the gateway to a second client for message-monitoring and forwarding across 1,000+ daily messages.
- **MAGe — Multi-Agent Media Generation Engine**
 - Hierarchical agent pipeline (creative director → scriptwriter → reference selector → generator → reviewer) producing on-brand video ads from a **Playwright**-scraped brand profile, using Google Veo with frame-carry continuity and FFmpeg assembly; tracks per-session API cost via Python **contextvars** across concurrent async workers.
- **Privacy-First On-Premise RAG (A2A Protocol)**
 - Inverted RAG topology for compliance-sensitive clients — the reasoning agent deploys *onto client infrastructure* over an **Agent-to-Agent protocol**, queryable by any external agent with **zero data exfiltration**; hybrid dense-vector and keyword index for mixed document types.

Quantashift Consultancy Services (contracted to MGS Technology)

Freelance Voice AI Consultant

Jan. 2026 – Mar. 2026

Remote (Pune, India)

- **Hireups — Real-Time AI Interviewer with Photorealistic Video Avatar** Hireups
 - Brought in after the internal team was blocked for months on LiveKit real-time *video*; delivered a **live photorealistic avatar interviewer** on Gemini's realtime audio-native model with sub-second, interruptible multilingual dialogue (English/Arabic in production).
 - Shipped **three production deployments** — GCP VM, Cloud Run (non-trivial: LiveKit workers require persistent connections), and the client's private server, which needed a *custom LiveKit build* to satisfy ZeroSSL on an untrusted IP range.
 - Rearchitected a failing WebRTC monolith into scalable GCP microservices; added a **multi-LLM failover mesh** (Gemini/Groq/Ultravox) for uninterrupted sessions, and moved proctoring inference in-browser (TensorFlow.js, MediaPipe, COCO-SSD) to eliminate server video compute.

Technical Skills

AI & Agents: Google ADK, LangGraph, LiteLLM, LightRAG, Multi-Agent Systems, A2A Protocol, RAG, Vector DBs

Languages & Cloud: Python (AsyncIO), Go, SQL, Docker, GCP (Cloud Run, Compute Engine, Vertex AI), DigitalOcean

Voice & Real-Time Systems: LiveKit (WebRTC), SIP/VoIP telephony, WebSockets, Gemini Live API, streaming STT/TTS

Backend & DevOps: FastAPI, REST APIs, OAuth 2.0, Redis, NGINX, GitHub Actions (CI/CD), Fail2Ban

Publication

A Demand-Driven Software Approach with Dynamic Micro-Expiry and Just-in-Time Processing to Reduce Platelet Wastage in Blood Banks

IEEE IC2PCT 2026, pp. 978–983 · [IEEE Xplore](#)

- Co-author — led problem formulation, forecasting-model development (SARIMA/XGBoost; SARIMA lowest MAE at 5.85), and simulation. Reduced simulated wastage **11.2%** → **2.5%** while holding **99.1%** fulfillment, statistically validated across 30 iterations (paired t-tests).

Projects

Fraud Detection System

[GitHub](#)

Infosys Springboard, ML Internship · ARIMA/Prophet, XGBoost

- Built a hybrid fraud-detection pipeline on a highly imbalanced **400+ feature** dataset (ARIMA/Prophet + XGBoost), achieving **96% fraud recall** with an optimised F1-score.

AnyAssist — Multi-Tenant RAG Platform

[AnyAssist](#)

GCP, LiteLLM, LightRAG, FastAPI, Google ADK

- Self-service “chat with your data” SaaS with strict per-tenant vector isolation; LiteLLM gateway handles cross-provider routing, rate limiting, and per-tenant cost tracking.

UniBias — Live Attention Tracker

[UniBias](#)

Vision Transformers, FastAPI, OpenCV

- Privacy-first tracker analysing periodic webcam and screen frames to detect distraction and trigger alerts, with no recording.

Education & Extracurricular

Ajay Kumar Garg Engineering College (AKGEC)

B.Tech in Computer Science · CGPA: 8.0

Ghaziabad, India

- **Coordinator, Cloud Computing Cell:** organised ML and cloud workshops for 200+ students.
- **Coordinator, Centre of Metaverse:** led “Rescue X”, a VR first-responder platform, to **Top 5** at the National IDE Boot-camp 2025.

Certifications

- Building AI Voice Agents for Production, LiveKit / DeepLearning.AI (Sep. 2025) · Foundations of Responsible AI (Sep. 2025)
- Complete Data Science, ML, DL & NLP Bootcamp, Udemy (Jul. 2025) · Fundamentals of Deep Learning, NVIDIA DLI (Mar. 2025)